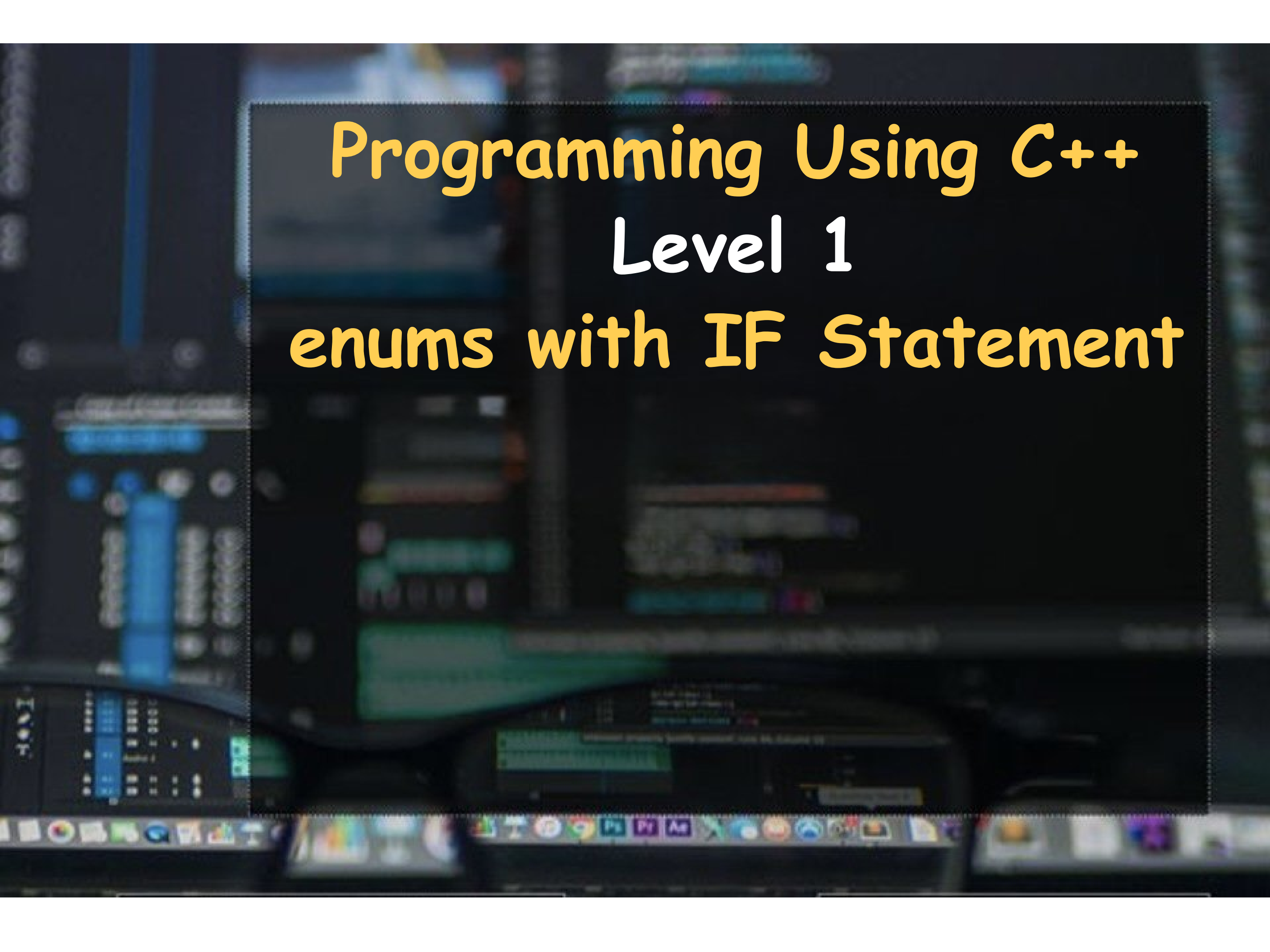
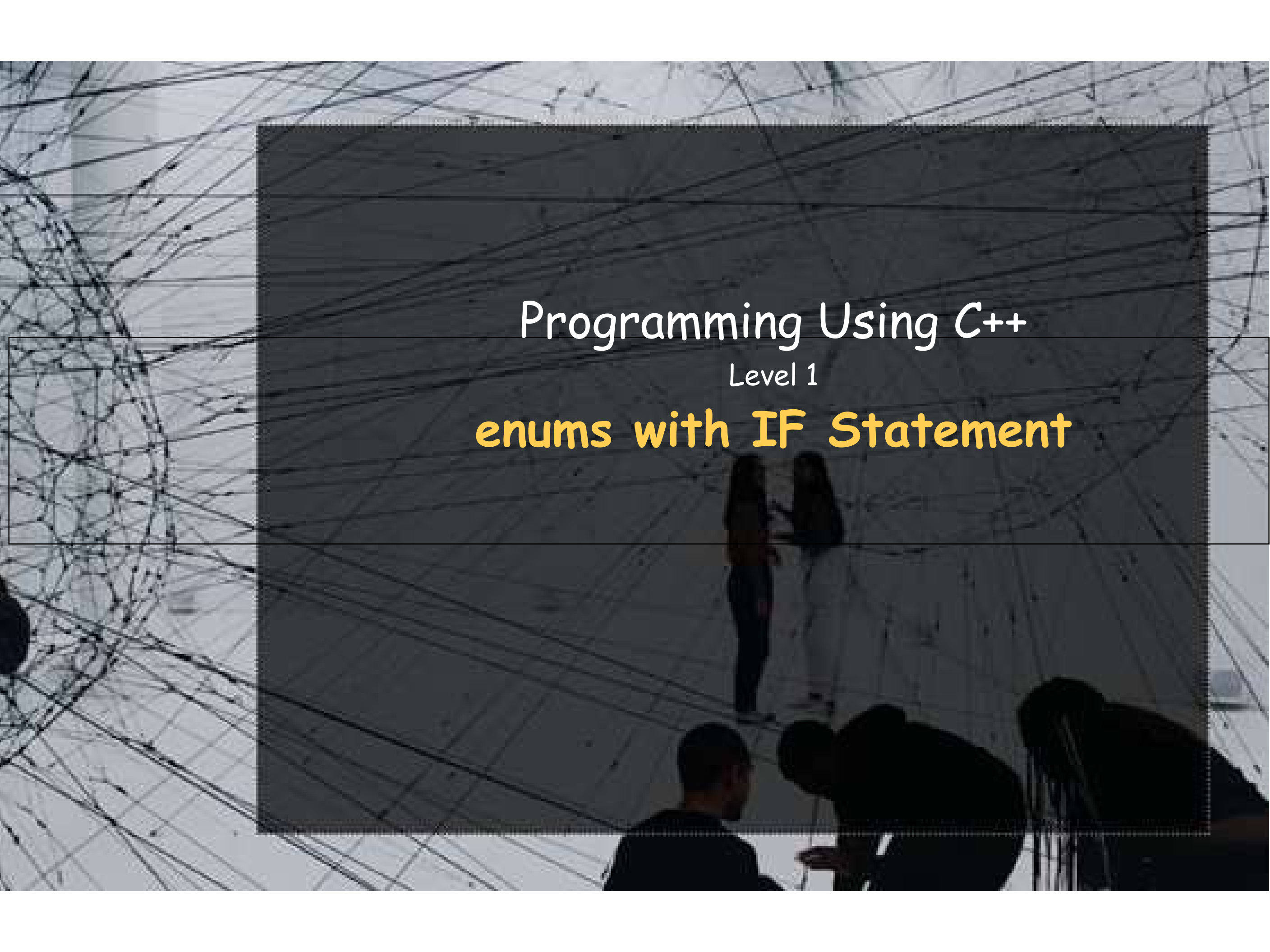




Programming Using C++ Level 1 enums with IF Statement



Programming Using C++

Level 1

enums with IF Statement

Definition

◆ This program defines an enum called `enSceenColor` with four color options.

 `Red = 1`

 `Green = 2`

 `Blue = 3`

 `Yellow = 4`

◆ The user enters a number, and `static_cast` converts that number to the `enSceenColor` enum type.

◆ `ChangeBackgroundColor()` then uses if and else if statements to determine which color was selected.

Explanation

- ◆ This program defines `MySumFunction()` with two parameters: `Num1` and `Num2`.
- ◆ The function receives two numbers, adds them together, and returns the result.
- ◆ The values entered by the user are passed to the function when `MySumFunction(Num1, Num2)` is called.

☐ Steps in main()

☐ Define the enum

◆ Create `enScreenColor` with Red, Green, Blue, and Yellow.

☐ Display the menu

◆ Show the available colors and their corresponding numbers.

☐ Read the user's choice

◆ Store the entered number in `userColor`.

☐ Convert the value

◆ Use `static_cast<enScreenColor>` to convert the integer into an enum value.

☐ Pass the enum to the function

◆ Send the converted value to `ChangeBackgroundColor()`.

☐ Check the selected color

◆ Use `if` and `else if` to compare the enum value with each color.

☐ Change the console color

◆ Execute the corresponding system command and display the selected color.

☐ Handle invalid choices

◆ Use `else` when the selected value does not match any defined color.

💡 Example

◆ If the user enters:

3

◆ The value is converted to:

enScreenColor::Blue

◆ The program executes the Blue condition and displays:

You chose Blue color.

➡ Result

◆ User enters 1 → Red

◆ User enters 2 → Green

◆ User enters 3 → Blue

◆ User enters 4 → Yellow

◆ Any other number → Invalid color choice.

Note

- ◆ Red, Green, Blue, and Yellow are enum members.
- ◆ The enum values are explicitly assigned from 1 to 4.
- ◆ `static_cast<enSceenColor>(userColor)` converts the integer entered by the user into the enum type.
- ◆ `system("Color 4F")`, for example, is a Windows console command used to change the console colors.

⊘ Warning

- ◆ `static_cast` does not validate whether the number corresponds to a defined enum member.
- ◆ For example, 10 can be converted to `enScreenColor` even though 10 is not one of the defined colors. The `else` statement handles this case.



Comparison

Enum

- ✓ Gives meaningful names to related constant values.
- ✓ Makes conditions easier to understand.
- ✓ Groups related choices into one type.

Integer

- ✓ Stores numeric values directly.
- ✓ Can represent any integer, including values that are not valid color choices.
- ✓ Is less descriptive than an enum.

✦✦ Summary

- ✓ Enums provide meaningful names for a group of related values.
- ✓ if and else if can be used to handle different enum choices.
- ✓ static_cast can convert the user's integer input into the enum type.

Thank you 😊